

HLA-ProtBERT: Multi-Encoder HLA Analysis

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National Marrow Donor Program

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- Production-ready pipeline with 5 encoders (ProtBERT, ESM-2, ProtT5, Ankh base/large)
- Caches embeddings/plots for 26k IMGT/HLA alleles (latest snapshot)
- Class I focus: 17,279 A/B/C alleles clustered across PCA, t-SNE, UMAP
- Resumable CLI scripts, GPU-aware, SSL bypass + HF token support

Benefits

- **Mgmt:** Efficient analysis pipeline
- **Clinical:** Better matching insights
- **Tech:** Novel functional view

- 1 HLA Analysis Challenge
- 2 AI-Based Approach
- 3 Key Findings
- 4 Applications

Content for:

- Management
- Clinical Scientists
- Bioinformaticians

- Thousands of alleles with critical immune function
- Sequence similarity is not equivalent to functional similarity
- Growing database makes manual analysis impossible

Clinical: Better transplant matching

Mgmt: Efficiency gains

Tech: Novel computational approach

Key Takeaway

HLA complexity requires advanced methods beyond simple sequence comparison.

HLA-ProtBERT: Multi-Encoder Pipeline

- Encoders: ProtBERT (768), ESM-2 (1280), ProtT5 (1024), Ankh base (768), Ankh large (1536)
- Shared encoder base: caching, locus filtering, SSL toggle, optional HF token
- GPU auto-detection; batch encoding to handle full IMGT/HLA catalog
- Pipeline script orchestrates downloads, embedding generation, and locus plots

Clinical: Finds functionally similar HLAs

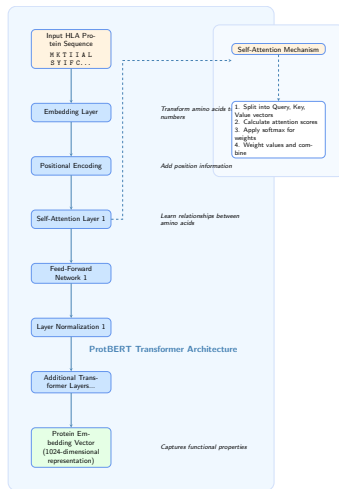
Mgmt: Leverages existing AI advances

Tech: Transfer learning from vast protein data

Key Takeaway

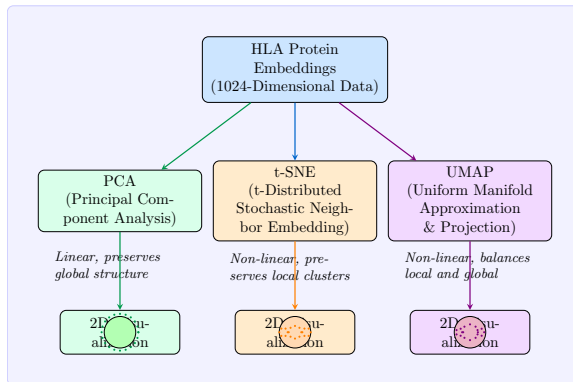
ProtBERT understands protein "language" beyond simple sequence similarity.

How It Works



- Processes HLA sequence while considering all amino acid relationships
- Multiple layers extract increasingly complex patterns

From Complex to Simple



Dimensionality Reduction Techniques

PCA

Global overview

t-SNE

Local clusters

UMAP

Balanced view

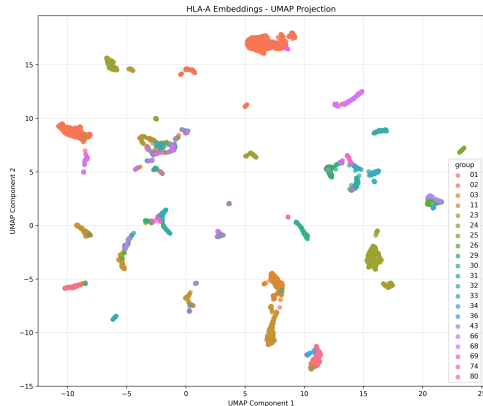
- 17,279 Class I alleles analyzed (ProtBERT embeddings):
 - 5,489 HLA-A
 - 6,584 HLA-B
 - 5,206 HLA-C
- Additional Class II data available (DRB1, DQB1, DPB1, etc.)
- Clear clustering by functional groups across PCA/t-SNE/UMAP

Clinical: Reveals functional similarities

Mgmt: Demonstrates efficiency at scale

Tech: Effective protein embedding

HLA-A Visualization



- Clear separation between major allele families
- A*01, A*02, A*03 groups form discrete clusters

HLA-A: Multiple Perspectives

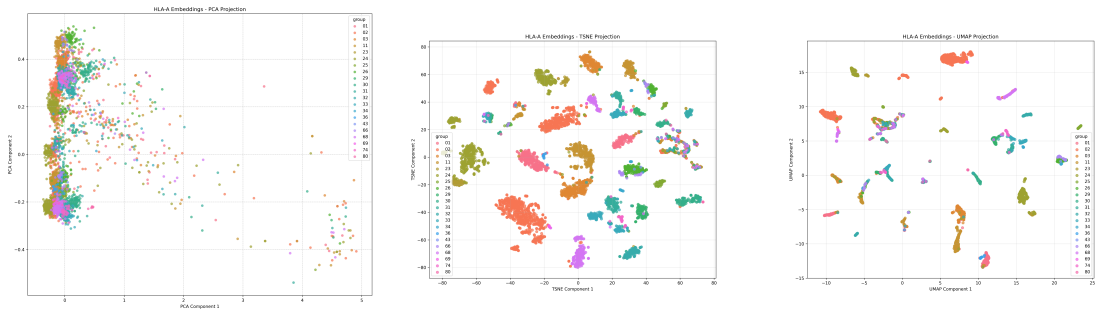
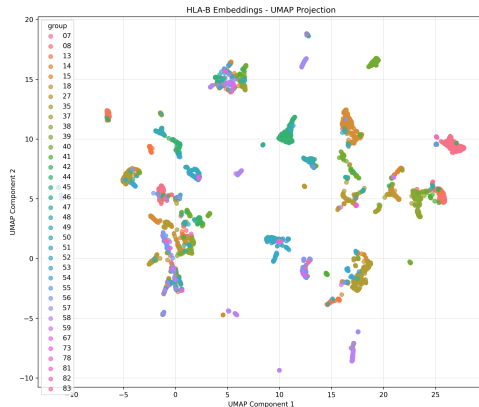


Figure: PCA (left), t-SNE (middle), UMAP (right)

Clinical: Helps identify potential cross-reactive epitopes

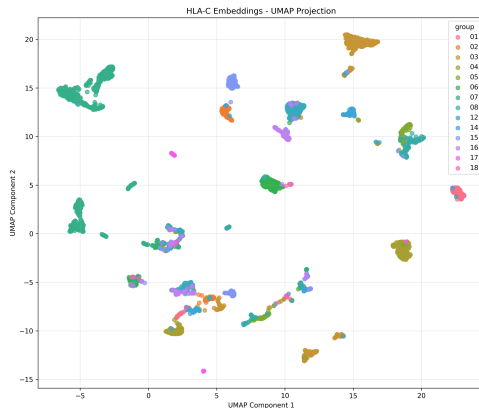
Tech: Different methods reveal different relationship aspects

HLA-B: Most Diverse



- Most diverse locus (6,526 alleles)
- Complex clustering reflects greater polymorphism

HLA-C: Distinct Patterns



- More compact clusters than A and B loci
- Clearer separation between major groups

Clinical

- **Clinical:** Better matching prediction
- **Clinical:** Novel allele classification
- **Clinical:** Cross-reactivity assessment

Research & Technical

- **Tech:** Structure-function modeling
- **Mgmt:** Efficient analysis workflow
- **Tech:** Evolutionary analysis

Understanding functional relationships between HLA alleles has direct applications in transplantation, disease association, and drug response prediction.

Technical Advancements

- HLA-specific training
- Structural information integration
- Multi-modal models

Data Integration

- Clinical outcome correlation
- Population genetics
- Cross-species analysis

Potential Impact

- **Clinical:** Improved virtual crossmatching
- **Mgmt:** Better matching algorithms
- **Tech:** Novel evolutionary insights

Key Takeaways

- 1 Multi-encoder embeddings reveal functional HLA relationships
- 2 17,279 Class I alleles analyzed from the Dec 2025 IMGT/HLA snapshot
- 3 Clear functional clustering across PCA, t-SNE, and UMAP
- 4 Resumable pipeline caches embeddings for ProtBERT, ESM-2, ProtT5, Ankh
- 5 Applications span clinical, research, and production workflows

Clinical: Explore matching prediction applications

Mgmt: Consider workflow integration

Tech: Explore additional models

Model Specifications

- ProtBERT: 420M params, 768-dim
- ESM-2 (650M): 1280-dim
- ProtT5 XL: 1.3B params, 1024-dim
- Ankh Base/Large: 50M/650M, 768/1536-dim
- Ensemble-ready; embeddings cached under `data/embeddings/`

Dimensionality Reduction Parameters

- **UMAP:** `n_neighbors=15`, `min_dist=0.1`
- **t-SNE:** `perplexity=30`, `learning_rate=200`
- **PCA:** `n_components=2`

Implementation

- `run_complete_pipeline_all_encoders.sh` orchestrates download & encoding
- Hugging Face downloads with optional HF token; `--disable-ssl-verify` flag
- GPU auto-detection; batch encoding defaults to 8
- Caching + resumable runs for embeddings and locus plots
- Locus analysis via `scripts/run_locus_analysis.py` (PCA/t-SNE/UMAP)